

Bridging Kolmogorov Complexity and Deep Learning

Shaw, Cohan, Eisenstein, Toutanova
Analyse: Firsov Sergey

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Introduction: Why MDL and Kolmogorov Complexity

MDL intuition

Learning is viewed as data compression: the best model minimizes the total length of the model description plus the description of the data encoded with that model.

Key idea

Any predictive regularity is also useful for compression, and vice versa. Predictive power and compressibility are two sides of the same coin.

Gap

For neural networks, measuring the “complexity of the function” is nontrivial, unlike counting the length of a discrete program. This creates a gap between algorithmic information theory and practical training objectives.

Goal of the paper

Establish asymptotically optimal description-length objectives for Transformers and provide a tractable, differentiable variational objective that can be optimized in practice.

Contributions and Why They Matter

1. Universal two-part codes for probabilistic models: their minimum is optimal up to an additive constant compared to any other two-part code.
2. Asymptotically optimal families of codes for Transformers: as resources grow, the minimum approaches the MDL optimum (up to a constant).
3. A tractable, differentiable variational objective using adaptive Gaussian mixture priors, enabling gradient-based optimization.
4. Empirical analysis: the objective selects low-complexity, strongly generalizing solutions; however, standard optimizers often fail to find them from random initialization.

Kolmogorov Complexity and Resource-Bounded Approximations

Definitions

$K(x)$: length of the shortest program that prints x . $K(f)$: length of the shortest program computing a discrete function f .

Properties

Invariance theorem: values are machine-independent up to a constant.
Exact K is not computable due to the halting problem.

Approximations

$K_{T,R}(f)$ considers only programs halting within R_t steps and using at most R_s registers. As R grows, $K_{T,R}(f)$ decreases toward $K(f)$ (up to a constant).

Inequalities up to constants

Use $f(x) \leq g(x)$ if $\exists c$ independent of x with $f(x) \leq g(x) + c$. Write $f(x) \approx g(x)$ if both directions hold.

Problem Setting and Model Functions

Data and models

X : discrete countable set; Y : finite set. Dataset $D = (X \times Y)^*$ is a finite sequence of pairs (x_i, y_i) .

Model function

$f \in \mathcal{F}$ maps x to logits $f(x) \in \mathbb{Q}^{|Y|}$. The conditional distribution is $p(y|x; f) = \sigma(f(x))_y$ and $p(Y|X; f) = \prod_{(x,y)} p(y|x; f)$.

Goal

Identify encoding schemes and objectives that optimally compress Y given X by selecting $f \in \mathcal{F}$, in the MDL sense.

Two-Part Codes: Mechanism and Purpose

Mechanism

Step 1: transmit a hypothesis h (model parameters) at cost $-\log_2 \alpha(h)$.

Step 2: transmit labels Y given X and $m(h)$ at cost $-\log_2 p(Y|X; m(h))$.

$$L_M^{\text{two-part}}(Y|X; h) = -\log_2 \alpha(h) - \log_2 p(Y|X; m(h)).$$

Purpose

This objective enforces a balance: model complexity via $-\log_2 \alpha(h)$ vs. data fit via $-\log_2 p(\cdot)$. Minimizing over h yields the MDL trade-off and coincides with MAP estimation.

Universal Two-Part Codes: Definition, Why It Works

Definition

A two-part code M is universal if for any other two-part code M' ,
 $L_M^{*,\text{two-part}}(Y|X) + \leq L_{M'}^{*,\text{two-part}}(Y|X)$.

Why it works

Two conditions: (1) the model class is computationally universal; (2) the prior assigns at least one hypothesis a code length matching $K(f)$ up to a constant. Then any computable regularity can be efficiently represented.

Why we need it

Removes arbitrariness in prior choice and provides near-optimal compression guarantees (up to a constant), for any dataset.

Asymptotically Optimal Families of Two-Part Codes

Definition

A family $\{M_R\}$ is asymptotically optimal if

$$L_{M_R}^{*,\text{two-part}}(Y|X) + \leq C_{T,R}^{\text{two-part}}(Y|X) \text{ and}$$

$$\lim_{R_t, R_s \rightarrow \infty} L_{M_R}^{*,\text{two-part}}(Y|X) + = C_{\text{two-part}}(Y|X).$$

Meaning

Provides a computable path to approach the ideal MDL bound by increasing resources, ensuring monotone improvement (up to a constant).

Connecting to Transformers: Idea, How, and Why

Idea

Treat Transformers as a resource-parameterized family: layers $\sim R_t$, context window $\sim R_s$. Use ALTA to map a program z to weights h_z so that $m(h_z) = f_T^z$.

How

Store z in prompt embeddings; configure attention/MLP as an interpreter of a prefix Turing machine. This yields a correspondence between programs and Transformer parameters.

Why and what for

Assign the prior so that $-\log_2 \alpha(h_z) = |z| \approx K(f_T^z)$. As resources grow, the two-part code for Transformers becomes asymptotically optimal, approaching the MDL bound.

Practical Limitations and the Need for Variational Objectives

Limitations

Rigid priors are non-differentiable and assign zero mass to most parameters; literal Turing emulation is inefficient for real tasks.

Need

Use flexible, differentiable codes compatible with standard optimizers, while retaining asymptotic guarantees as upper bounds.

Variational Codes: Formula, Intuition, and Quasi-Universality

Variational code

Posterior $\beta(h; \varphi)$ over hypotheses with prior $\alpha(h)$.

$$L_M^{\text{var}}(Y|X, \varphi) = \mathbb{E}_{h \sim \beta} [-\log_2 \alpha(h) + \log_2 \beta(h; \varphi) - \log_2 p(Y|X; m(h))] = KL[\beta \| \alpha].$$

Quasi-universality

$$K(Y|X) + \leq C_{\text{bayes}}(Y|X) + \leq L_M^{*, \text{var}}(Y|X) + \leq C_{\text{two-part}}(Y|X).$$

Adaptive GMM priors

Products of independent GMMs per weight groups: soft quantization, reparameterization-based training, and an asymptotically optimal family for Transformers.

Experiments (1/2): Purpose and Setup

Task

Parity: given a binary sequence, predict whether the number of ones is odd or even. A canonical stress test for length generalization in Transformers.

Purpose

Disentangle whether the variational objective fails to select compressible, generalizing solutions, or whether optimization from random initialization fails to find them.

Method

Use ALTA to construct a manual solution that fits training data and has low variational codelength (an upper bound). Then train from random initialization with the same objective and compare to an MLE baseline.

Experiments (2/2): Findings and Implications

Findings

Manual initialization under the variational objective achieves low codelength and perfect OOD length generalization; random initialization fails to match it; MLE fits train but fails OOD.

Analysis

From random starts, priors/posteriors tend to collapse into unimodal distributions, inflating KL and harming compression. The manual solution is multimodal with low-variance components, yielding much smaller KL.

Implications

The objective is sound; optimization is the bottleneck. Methods that prevent collapse or alternative asymptotically (quasi-)optimal families are promising directions.

Conclusion and Future Work

Summary

A theoretical-practical bridge from Kolmogorov complexity to training objectives for Transformers: universal/asymptotically optimal two-part codes and a differentiable variational objective with adaptive GMM priors.

Main challenge

Optimization from random initialization does not reliably find low-complexity solutions selected by the objective.

Future work

Design optimizers or code families preventing posterior collapse; extend to other architectures (decoders, CoT) and resource scaling dimensions; integrate external tools.